

D 1 A

$$y = \ln(x^2 + 3x + 4)$$

$$y' = \frac{1}{x^2 + 3x + 4} \cdot 2x + 3 = \frac{2x + 3}{x^2 + 3x + 4}$$

$$\frac{2x + 3}{x^2 + 3x + 4} \geq 0 \quad \left\{ \begin{array}{l} 2x + 3 \geq 0 \\ \forall x \end{array} \right\} \left\{ \begin{array}{l} 2x \geq -3 \\ \forall x \end{array} \right\} \left\{ \begin{array}{l} x \geq -3/2 \\ \forall x \end{array} \right.$$

$$f' > 0 \text{ pour } x \in (-3/2, +\infty)$$

D 1 B

$$y = \arctan(x^2 - 5x + 7)$$

$$y' = \frac{1}{(x^2 - 5x + 7)^2 + 1} \cdot 2x - 5 = \frac{2x - 5}{(x^2 - 5x + 7)^2 + 1}$$

$$\frac{2x - 5}{(x^2 - 5x + 7)^2 + 1} \geq 0 \quad \left\{ \begin{array}{l} 2x - 5 \geq 0 \\ (x^2 - 5x + 7)^2 + 1 \geq 0 \end{array} \right\} \left\{ \begin{array}{l} x \geq 5/2 \\ \forall x \end{array} \right.$$

$$f' \geq 0 \text{ pour } (-5/2, +\infty)$$

D 3 A

$$y = e^{x - x^2}$$

$$y(0) = e^{0 - 0} = 1$$

$$y' = e^{x - x^2} (1 - 2x) = e^{x - x^2} - 2x e^{x - x^2}$$

$$y'(0) = e^{0 - 0} - 2 \cdot 0 \cdot e^{0 - 0} = 1$$

$$y'' = [(e^{x - x^2})(1 - 2x)](1 - 2x) + (e^{x - x^2})(-2)$$

$$(e^{x - x^2} - 2x e^{x - x^2})(1 - 2x) - 2 e^{x - x^2}$$

$$e^{x - x^2} - 2x e^{x - x^2} - 2x e^{x - x^2} + 4x e^{x - x^2} - 2 e^{x - x^2} \quad y''(0) = e^{0 - 0} - 2 \cdot 0 \cdot e^{0 - 0} - 2 \cdot 0 \cdot e^{0 - 0} + 4 \cdot 0 \cdot e^{0 - 0} - 2 e^{0 - 0} = -1$$

$$1 + x - \frac{x^2}{2}$$

D 3 B

$$y = e^{-x+x^2} \quad y(0) = e^{-0+0} = 1$$

$$y' = e^{-x+x^2} (-1+2x) \quad y'(0) = e^{-0+0} (-1+2 \cdot 0) = -1$$

$$y'' = [(e^{-x+x^2})(-1+2x)](-1+2x) + (e^{-x+x^2})(+2) \quad y''(0) = [(e^{-0+0})(-1)](-1) + (e^0)(+2) = 1 + 2 = 3$$

$$1 - x + \frac{3x^2}{2}$$

D 4 A

$$\int_0^{e^2} \frac{\ln^2(x)}{x} = \left[\frac{\ln^3(x)}{3} \right]_0^{e^2} = \left(\frac{\ln^3(e^2)}{3} \right) = \frac{2 \ln^3(e)}{3} = \frac{8}{3}$$

$$\int \frac{\ln^2(x)}{x} = \int \frac{x \cdot \ln^2(x)}{x} = \int y^2 = \frac{y^3}{3} = \frac{\ln^3(x)}{3} + C$$

$$y = \ln(x)$$

$$dy = \frac{dx}{x}$$

$$x \, dy = dx$$

D 4 B

$$\int_0^1 \frac{\arctan(x)}{1+x^2} = \left[\frac{\arctan^2(x)}{2} \right]_0^1 = \left(\frac{\arctan^2(1)}{2} \right) - 0 = \left(\frac{\pi}{4} \right)^2 \cdot 2 = \frac{\pi^2}{16} \cdot \frac{1}{2} = \frac{\pi^2}{32}$$

$$\int \frac{\arctan(x)}{1+x^2} = \int \frac{\cancel{(1+x^2)} \arctan(x)}{\cancel{1+x^2}} = \int y = \frac{y^2}{2} = \frac{\arctan^2(x)}{2} + C$$

$$y = \arctan(x)$$

$$dy = \frac{dx}{1+x^2}$$

$$(1+x^2) \, dy = dx$$

D 7 A

$$\lim_{n \rightarrow +\infty} \frac{n!}{n^{n/3}} = +\infty \quad \text{per le infinite}$$

$$\ln(n) \ll n^e \ll e^n \ll n! \ll n^n$$

D 7 B

$$\lim_{n \rightarrow +\infty} \frac{n!}{n^{3n}} = 0 \quad \text{per le infinite}$$

$$\ln(n) \ll n^e \ll e^n \ll n! \ll n^n$$

D 8 A

$$\sum_{n=1}^{+\infty} \frac{(n+2)^n}{(3n)^n} \quad \frac{\sqrt[n]{(n+2)^n}}{\sqrt[n]{(3n)^n}} = \frac{n+2}{3n} \sim \frac{n}{3n} = \frac{1}{3} \quad \frac{1}{3} < 1 \quad \sum a_n \text{ converge}$$

D 8 B

$$\sum_{n=1}^{+\infty} \frac{(4n+2)^n}{(3n)^n} \quad \frac{\sqrt[n]{(4n+2)^n}}{\sqrt[n]{(3n)^n}} = \frac{4n+2}{3n} \sim \frac{4n}{3n} = \frac{4}{3} \quad \frac{4}{3} > 1 \quad \sum a_n \text{ diverge}$$

D 6 B

$$f = \begin{cases} 2^{-x} & -2 < x < 1 \\ 1/2x & -1 \leq x \leq 2 \end{cases}$$

$$f = 2^{-x} \quad (-1) = \ln(2) \cdot 2^{-x} \quad (-1) = -\frac{\ln(2)}{2^x}$$

$$-\frac{\ln(2)}{2^x} > 0 \quad \left\{ \begin{array}{l} -\ln(2) > 0 \\ 2^x > 0 \end{array} \right\} \forall x \Rightarrow \emptyset \quad f \text{ è sempre decrescente}$$

$$f = \frac{1}{2x}$$

$$f' = \frac{-(1)(2)}{(2x)^2} = \frac{-2}{4x^2} = -\frac{1}{2x^2}$$

$$-\frac{1}{2x^2} > 0 \quad \left\{ \begin{array}{l} -1 > 0 \\ 2x^2 > 0 \end{array} \right\} \forall x \Rightarrow \emptyset \quad f \text{ è sempre decrescente}$$

D 6 A

$$y = \begin{cases} 2^{-x} & -2 < x < 1 \\ 1/x & 1 \leq x < 2 \end{cases}$$

$$f = 2^{-x}$$

$$f' = \ln(2) \cdot 2^{-x} \cdot -1 = -\frac{\ln(2)}{2^x}$$

f' è sempre decrescente

$$g = 1/x$$

$$g' = \frac{-(1)(1)}{x^2} = -\frac{1}{x^2}$$

$$-\frac{1}{x^2} > 0 \quad \begin{cases} -1 > 0 \\ x^2 > 0 \end{cases} \quad \begin{cases} \emptyset \\ \forall x \end{cases}$$

$$\lim_{x \rightarrow 1^-} 2^{-x} = \frac{1}{2} \quad \text{la funzione ha una discontinuità}$$

$$\lim_{x \rightarrow 1^+} 1/x = 1$$

La funzione ha immagine $(1/2, 1)$

Ex 2A

$$y = \begin{cases} x^2 - x & x \leq 1 \\ \frac{e^x - e}{(x-1)^2} & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} 1 - 1 = 0$$

$$\lim_{x \rightarrow 1^+} \frac{e - e}{0} = \frac{0}{0} \stackrel{H}{=} \frac{e}{1 - 2} = \frac{e}{-1} = -e$$

La funzione ha una discontinuità di 2° tipo

Ex 2B

$$y = \begin{cases} x^2 - \frac{x}{2} & x \leq 1 \\ \frac{e^x - e}{2(x-1)^2} & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} 1 - \frac{1}{2} = \frac{2 - 1}{2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 1^+} \frac{e - e}{2(0)^2} = \frac{0}{0} \stackrel{H}{=} \frac{e}{4 - 4} = +\infty$$

$$2(x-1)^2 = (2)(2(x-1)(1)) = (2)(2x-2) = 4x-4$$

La funzione ha una discontinuità di 2° tipo in $x=1$

Eg 1

$$y = \begin{cases} \arctan\left(\frac{x+1}{x^2}\right) \\ e & x=0 \end{cases}$$

1) $f(x)$ è continua in $x=0$ se $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\lim_{x \rightarrow 0} \arctan\left(\frac{x+1}{x^2}\right) = \lim_{y \rightarrow +\infty} \arctan(y) = \pi/2$$

$$y = \frac{x+1}{x^2}$$

$$y \rightarrow +\infty$$

$$x \rightarrow 0$$

$\pi/2 \neq \pi/2$ cioè che $f(x)$ non è continua in $x=0$

2)

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \frac{\pi/2 - \pi/2}{0} = \frac{0}{0} \stackrel{H}{=} - \frac{h^2 + 2h}{h^4 + (h+1)^2} = \frac{0 + 2 \cdot 0}{0 + (0+1)^2} = \frac{0}{1} = 0$$

Per il rapporto in numeratore è derivabile in $x=0$